Evaluation of Digital Technology Applications in Anatomy Education and Impact on Academic Performance of Students

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ABSTRACT

Anatomy education involves the study of the structure and function of the body, essential for health and medical professions. This study investigates the socio-demographic characteristics, digital technology usage, perceived effectiveness, and impact on academic performance among undergraduate students in Anatomy Education. A quantitative analysis was conducted with a sample predominantly aged 18–22 years (65.9%) and a significant female representation (69.3%). Findings reveal that digital tools, particularly educational videos (48.9%) and 3D anatomy models (39.8%), are frequently utilized to enhance learning experiences, with 28.4% of participants reporting frequent usage. A substantial proportion (45.3%) found digital tools helpful in understanding anatomical concepts, while nearly half (48.8%) reported significant improvements in academic performance attributed to these resources. The study highlights the transformative potential of digital technologies in anatomy education, emphasizing their role in fostering engagement, knowledge retention, and practical application of learning. Recommendations include integrating digital tools into curricula, enhancing faculty training, and promoting technological applications in health science education. Overall, the findings underscore the necessity for educational institutions to adapt to technological advancements to enrich student learning and prepare them for future professional challenges.

Key words: Digital tools, technological applications, anatomy education, academic performance, undergraduate students.

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1. INTRODUCTION

Anatomy education has transformed over time from traditional blackboard teaching to modern methods like overhead projectors, PowerPoint presentations, 3D models, virtual dissection, and online remote learning using 3D virtual dissection. Understanding human anatomy is fundamental in medical practice, and it's the responsibility of anatomy teachers to ensure their students become competent clinicians [1]. Modern teachers not only impart knowledge but also facilitate the learning process by leveraging on innovative technologies. While the use of educational technologies is not new, their necessity has grown significantly in the current landscape. Therefore, anatomy teachers must stay updated with advanced technologies in medical education. For students, their learning habits have evolved, and they now embrace various technologies, such as YouTube videos, learning apps, and online workshops, to refine their knowledge and dissection skills [2]. New tools like anatomy studios, virtual dissectors, simulation labs, and radiology equipment have become key components of anatomy education. Furthermore, the emergence of endoscopic anatomy presents a fresh perspective on the body, going beyond

anatomical positions [3]. Given these evolving trends, we aim to explore technologies advancement in delivering anatomy instruction.

Anatomy education is passing through a paradigm shift, online web-based remote learning, and portable network devices are becoming a fundamental component of the learning environment in anatomy education, having an impact on design, and re-shaping the anatomy curriculum. In today's scenario, a vast wealth of web-based information in the form of automated learning-system or interactive programs is available at the student's fingertips which can be used as a source of self-directed learning. These days, vast information are available online, starting from lecture schedule, ready to see presentations, examination schedule, results of assessment by one click [4].

Courses are predominantly delivered online via zoom meetings, google classroom, Microsoft teams, etc. Although formal lectures are still predominantly used for anatomy education, these new communication and information technologies can enhance the teaching and learning environment. They have the capacity to facilitate better integration between learning and problem-solving approaches, as they impact large amounts of factual information in an organized fashion in the digitalized form to make conventional didactic lectures more interesting. Although web-based platforms are popular among students, they need to be vigilant [4].

In medical curricula, Anatomy is considered one of the keystone courses. The prerequisite of a deep understanding of human anatomy is essential for safe clinical practice [1]. Without it, regardless of the area of healthcare, practitioners can experience difficulties in performing investigations effectively as they require knowledge of the precise locations of organs and tissues [11]. Traditional cadaver-based training does not tackle the critical need for growing healthcare advancements. Careful dissection is time-consuming, and access to dissection laboratories is generally limited [8]. There is also a high cost associated with maintaining a dissection laboratory. The use of two-dimensional images limits the ability to be creative and visual, thus limiting the correlation with clinical sciences. Textual explanation of three-dimensional structures is often inefficient for students to comprehend.

Digital technology has revolutionized numerous fields, including education, where its integration has brought about significant changes in teaching methodologies and learning outcomes. In anatomy education, digital tools such as virtual reality (VR), augmented reality (AR), interactive software, and 3D anatomical models have become increasingly prevalent. These technologies promise to enhance learning experiences, improve comprehension of complex anatomical structures, and potentially lead to better educational outcomes. This review evaluates the impact of digital technology on anatomy education, focusing on its effectiveness, benefits, and challenges, as well as its overall effect on students performance.

Digital tools also contribute to increased student engagement and motivation. Interactive software and VR environments make learning more dynamic and engaging compared to traditional methods. A study by [2] found that students who used AR and VR tools in anatomy education reported higher levels of motivation and interest in the subject. The novelty and interactivity of these tools create a more stimulating learning environment, which can lead to improved academic performance [2].

The aim of this study is to examine Digital Technology Applications in Anatomy Education and its impact on Academic Performance of Students. Specific objective is to determine the Digital Technology Usage and Perceived Effectiveness of the impact of digital Technologies on Academic Performance of Students. As medical knowledge grows and learners' expectations

shift towards technology-rich environments, digital tools like virtual reality (VR), augmented reality (AR), and 3D models offer promising alternatives. These technologies can enhance spatial comprehension, alleviate practical constraints like high costs and limited access to cadavers [8], and align with modern students' digital habits. By examining the effectiveness of these tools, the study aims to modulate curriculum development and support evidence-based practices in medical education [9]. The findings will provide insights into integrating digital technologies in anatomy teaching, potentially leading to more engaging and effective learning experiences [7].

2. MATERIALS AND METHODS

2.1. Study Design: A cross-sectional study design was used for this study with the target population being 100 to 400 Level students in the Department of Anatomy of Ambrose Alli University, Ekpoma.

2.2. Sampling technique and sample size: Simple random sampling technique was applied in selecting the participants.

2.3. Method of data collection: The primary instrument for data collection was a structured questionnaire that covered various aspects relating to the application of Digital technology in the study of Anatomy, such as the impact on anatomy education and academic performance of Students. The researcher made sure the participants satisfied the inclusion criteria before administering the questionnaire.

3.4. Ethical approval: Ethical approval was obtained from the Ambrose University ethical committee with approved number [043/25]. The respondents were informed that participation in the study is voluntary while they were given an assurance that information they provide will be treated confidentially. No personal identifiers, such as names, were recorded on the questionnaires. The information shared by respondents remained solely for research purpose.

3.5. Statistical analysis: Data was analyzed using IBM SPSS version 26.0. Descriptive statistics, including frequency and percentage distributions, to summarize socio-demographic characteristics, digital technology usage patterns, perceived effectiveness, and academic impacts. To assess associations between categorical variables, chi-square tests were performed. A level of $p<0.05$ was set to determine statistically significant differences.

3. RESULTS

Socio-Demographic Information: The study revealed key insights into the socio-demographic distribution of participants. A significant majority (65.9%) of respondents were aged between 18–22 years, with fewer participants aged 23–27 (33.0%) and 28–32 (1.1%) ($\chi^2 = 31.551$; $p = 0.000$). This age distribution reflects the typical demographic of undergraduate students in the studied population. Gender analysis showed a higher proportion of female participants (69.3%) compared to males (30.7%), with a statistically significant gender difference ($\chi^2 = 13.333$; $p = 0.000$). Most participants were in their 4th year (88.6%), with a smaller representation from the 2nd year (9.1%), 3rd year (1.1%), and 5th year (1.1%) ($\chi^2 = 137.580$; $p = 0.000$). This result highlights the academic distribution of the population, where more senior students are likely engaged in advanced academic tasks such as the use of digital tools for anatomy learning.

Digital Technology Usage: Patterns of digital technology usage revealed that participants frequently used digital tools for anatomy learning. A notable 28.4% reported frequent usage, while 23.9% used them always. In contrast, a small proportion (3.4%) never used these tools ($\chi^2 = 36.652$; $p = 0.000$), showing significant variation in usage frequency among participants.

Table 1: Socio-Demographic Information

Category	Frequency	Percentage	Chi-square	p-value
Age				
18-22	58	65.9		
23-27	29	33.0	31.551	0.000
28-32	1	1.1		
Gender				
Male	27	30.7	13.333	0.000
Female	61	69.3		
Year of Study				
2nd Year	8	9.1		
3rd Year	1	1.1	137.580	0.000
4th Year	78	88.6		
5th Year	1	1.1		

Table 2: Digital Technology Usage

Category	Frequency	Percentage	Chi-square	p-value
How often do you use digital technology tools?				
Never	3	3.4		
Rarely	13	14.8		
Occasionally	26	29.5	36.652	0.000
Frequently	25	28.4		
Always	21	23.9		
What digital technology tools do you use for learning anatomy? Choose all that apply.				
Virtual dissection	9	10.2		
3D anatomy models	35	39.8		
Educational videos/tutorial	43	48.9		
Online quizzes or interactive platforms	21	23.9	66.537	0.000
Digital textbooks or e-books	29	33		
Augmented reality	4	4.5		
Others	17	19.3		
How comfortable are you with using digital technology tools for anatomy learning?				
Slightly comfortable	10	11.4		
Neutral	26	29.5	12.114	0.007
Comfortable	31	35.2		
Very comfortable	21	23.9		
Do you access these digital tools primarily for self-study or part of a course curriculum?				
Self-study	38	43.2		
Part of a course curriculum	7	8	33.667	0.000
Both	43	48.9		
At what age do you start using digital technology?				
15-20	69	78.2	23.280	0.000
21-26	19	21.8		

When asked about specific tools used for anatomy learning, the most commonly mentioned were educational videos or tutorials (48.9%), 3D anatomy models (39.8%), and digital textbooks or e-books (33%). Other tools such as virtual dissection (10.2%) and augmented reality (4.5%) had relatively lower usage (χ^2 = 66.537; p = 0.000). These results indicate a preference for widely accessible and interactive digital tools that support visual and practical learning. Comfort levels with digital tools varied, with 35.2% reporting being comfortable and 23.9% very comfortable. However, a smaller group (11.4%) was only slightly comfortable (χ^2 = 12.114; p = 0.007). Most participants began using digital technology between the ages of 15–20 years (78.2%), with a smaller group (21.8%) starting later at 21–26 years (χ^2 = 23.280; p = 0.000).

Perceived Effectiveness: The perceived effectiveness of digital technology tools in anatomy education was overwhelmingly positive.

Table 3: Perceived Effectiveness

Category	Frequency	Percentage	Chi-square	p-value
Do you find digital technology applications helpful in understanding anatomical concept?				
Somewhat helpful	4	4.7		
Neutral	9	10.5	56.342	0.000
Helpful	40	45.3		
Very helpful	35	39.5		
How effective is digital technology tools compare to traditional learning methods?				
Much less effective	5	5.8		
Less effective	6	7		
About the same	14	16.3	77.213	0.000
More effective	44	50		
Much more effective	18	20.9		
Which aspects of anatomy learning did digital tools help you understand best?				
Structure and function of organs	40	45.9		
Spatial relationships between body parts	4	4.7		
Processes like circulation and respiration	6	7.1	84.612	0.000
Understanding 3D anatomy	25	28.2		
Others	12	14.1		
Have you noticed any improvement in your retention of anatomical knowledge after knowledge after using digital technology tools?				
Slightly improvement	16	18.6		
Moderate improvement	30	33.7	13.279	0.001
Significant improvement	42	47.7		

A significant proportion (45.3%) found digital tools helpful, while 39.5% rated them as very helpful in understanding anatomical concepts (χ^2 = 56.342; p = 0.000). When comparing digital tools to traditional methods, half of the respondents (50%) perceived digital tools as

more effective, and 20.9% considered them much more effective ($\chi^2 = 77.213$; $p = 0.000$). This underscores the transformative potential of digital technologies in anatomy education. Regarding specific learning outcomes, digital tools were most helpful in understanding the structure and function of organs (45.9%) and 3D anatomy (28.2%), while only 4.7% identified spatial relationships as a major benefit ($\chi^2 = 84.612$; $p = 0.000$). Moreover, nearly half of the participants (47.7%) reported significant improvement in knowledge retention after using digital tools ($\chi^2 = 13.279$; $p = 0.001$).

Table 4: Impact on Academic Performance

Category	Frequency	Percentage	Chi-square	p-value
Has the use of digital tools in anatomy education imparted your academic?				
Significantly decreased	1	1.1		
Slightly decreased	1	0.8		
No change	3	3.2	110.563	0.000
Slightly increased	40	45.3		
Significantly increased	43	48.8		
Which aspect of your academic performance has improved the most due to digital technology?				
Memorization of anatomical structures	11	12.8		
Understanding of complex systems	18	20.9		
Ability to apply knowledge in practical	25	27.9	52.419	0.000
Test scores/Exams	15	17		
Practical or laboratory skills	12	14		
Others	6	7		
Do you believe that using digital technology tools has contributed to higher engagement and motivation in your anatomy studies?				
Not at all	2	2.3		
To a small extent	9	10.3		
To some extent	45	50.6	74.516	0.000
To a great extent	32	36.8		
Do you feel that digital technology tools help you prepare more effectively for exams or theoretical assessment?				
Not effective	2	2.3		
Slightly effective	18	20.9		
Moderately effective	31	34.9	40.516	0.000
Very effective	37	41.9		

Impact on Academic Performance: Digital tools had a substantial impact on participants' academic performance. Nearly half (48.8%) reported a significant increase in their

academic achievements, while 45.3% noted slight improvements (χ^2 = 110.563; p = 0.000). In terms of specific academic improvements, the ability to apply knowledge in practical settings was the most enhanced area (27.9%), and followed by improved understanding of complex systems (20.9%). Other benefits included better memorization of anatomical structures (12.8%) and improved test scores (17%) (χ^2 = 52.419; p = 0.000). The tools also contributed to higher engagement and motivation, with 36.8% of participants stating they were engaged to a great extent and 50.6% to some extent (χ^2 = 74.516; p = 0.000). Similarly, 41.9% found these tools very effective in exam preparation (χ^2 = 40.516; p = 0.000).

4. DISCUSSION

The socio-demographic insights from this study highlight crucial trends in the profile of undergraduate students in Anatomy Education. A striking 65.9% of respondents fell within the age range of 18–22 years (Table 1), representing the typical demographic for undergraduate programs. This age distribution is significant as it indicates that younger students are at a pivotal point in their educational journeys, where foundational knowledge in anatomy can significantly shape their future professional competencies.

The predominance of female participants (69.3%) compared to males (30.7%) in Table 1 warrants further examination. This gender disparity may reflect broader trends in higher education, where fields such as health sciences often attract more female students. Previous studies have indicated that female students are more likely to enroll in programs that emphasize care and health, resulting in an increased female presence in medical courses [5].

Majority of participants were in their fourth year 88.6% (Table 1), suggesting that these students are likely familiar with the academic demands of their programs and the potential benefits of integrating digital tools into their learning processes. This familiarity can lead to more nuanced perspectives on the effectiveness of digital technologies. The reliance on digital tools for learning can facilitate deeper engagement with complex materials, aligning with the constructivist theories of learning where students build knowledge through active engagement [12].

The patterns of digital technology usage revealed a robust engagement with various educational tools, with 28.4% of participants frequently using these resources and 23.9% utilizing them consistently (Table 2). This engagement reflects a larger trend in education, where digital resources are becoming integral to teaching and learning.

The preferences expressed for specific tools-educational videos (48.9%), 3D anatomy models (39.8%), and digital textbooks (33%) in Table 2 underscore a significant shift towards visual and interactive learning methods. Furthermore, tools such as virtual dissections and augmented reality, although used less frequently, represent cutting-edge methodologies that can potentially revolutionize anatomy education. The lower engagement with these tools could indicate a need for greater exposure and training among students to leverage their full potential.

The comfort levels with digital tools varied among participants, with a majority 35.2% feeling comfortable and 23.9% feeling very comfortable (Table 2). This variance highlights a critical area for educational institutions to address: providing comprehensive training and support for all students, particularly those who may be less familiar or confident with digital technologies. The findings suggest that institutions should create an environment where all students feel equipped to utilize these tools effectively.

Furthermore, the starting age for digital technology use reveals that a significant majority (78.2%) began using these tools

between the ages of 15-20 (Table 2), indicating that early exposure to technology may foster a more intuitive use of digital resources in academic settings. This observation aligns with findings from Zhao *et al.* [13], who reported that early technology adoption positively influences academic performance. However, it also suggests that institutions should proactively introduce digital tools to students as early as possible in their academic careers, thereby nurturing a culture of technological proficiency.

When examining the perceived effectiveness of digital tools in anatomy education, the overwhelmingly positive responses, 45.3% found them helpful, 39.5% very helpful (Table 3), reflect a growing recognition of the transformative potential of digital resources in learning environments. The comparison between digital tools and traditional methods, where 50% of respondents deemed digital tools as more effective, emphasizes the changing landscape of educational preferences.

Participants also reported significant improvements in knowledge retention when utilizing digital tools, with 47.7% acknowledging a notable enhancement (Table 2). This suggests that digital tools not only facilitate immediate learning but also contribute to longer-term retention, a crucial aspect of anatomy education where foundational knowledge is built upon in subsequent learning experiences. The correlation between digital tool usage and improved academic performance further solidifies the argument for their integration into curricula.

The substantial impact of digital tools on academic performance, with nearly half of participants (48.8%) reporting significant improvements (Table 4), underscores the effectiveness of these technologies in enhancing educational outcomes. The benefits cited, particularly in practical application of knowledge and understanding complex systems, align with contemporary pedagogical theories that advocate for experiential and applied learning [3]. The improvement in memorization and test scores reflects a direct link between digital technology usage and academic success, suggesting that institutions must prioritize digital literacy and resource availability in their educational strategies.

Moreover, the enhanced engagement and motivation reported by participants underscore the role of digital tools in fostering a more stimulating learning environment. With 36.8% of respondents (Table 4), indicating that these tools contributed significantly to their engagement, it is evident that integrating technology can create a more dynamic and interactive educational experience. As education continues to evolve in response to technological advancements, it is crucial for institutions to embrace these changes and adapt their curricula accordingly.

5. CONCLUSION

This study highlights the significant role of digital technology in shaping the educational experiences of undergraduate anatomy students. The socio-demographic profile of the participants, along with their engagement with digital tools, brought into fore the necessity of incorporating innovative technologies into higher education curricula. The findings suggest that digital tools are not only enhancing student understanding of complex anatomical concepts but also contributing to improved academic performance and greater engagement in learning activities. The positive perceptions of digital technologies among students indicate a readiness to embrace new learning methods, reflecting broader trends in educational practices. This study provides a strong foundation for further exploration into the long-term impacts of digital technologies in education and their potential to transform teaching and learning methodologies.

Therefore digital technology continues to advance; its integration into anatomy education is not merely an enhancement but a necessity. This study's findings provide a clear

call to action for educational institutions to embrace these changes, ensuring that students are well-equipped for their future careers in health sciences and related fields.

RECOMMENDATIONS

It is important to create specialized training sessions and provide comprehensive resources for both educators and students, ensuring they are equipped to effectively integrate and utilize digital tools in anatomy education.
Regular evaluations should be implemented to assess the impact and effectiveness of digital technologies on enhancing learning outcomes in anatomy education, helping to refine and improve their use.

Adopting a hybrid learning model that blends traditional teaching methods with digital technologies can better accommodate various learning styles and improve overall educational effectiveness.

Cooperation between academic institutions and technology developers should be encouraged to create tailored digital solutions that meet the specific needs of anatomy education, ensuring tools are designed with the discipline's requirements in consideration.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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